# SQL Interview Preparation

Hindi + English Answers | Fresher Data Analyst | Concepts, examples, practical interview responses

Complete practice set: 40 questions with English answer, Hindi explanation, and examples where useful.

## How to use this PDF

First understand the Hindi explanation, then practice saying the English answer aloud. For code/formulas, explain what each part does. Adapt project claims to what you actually built and verified.

### *Q1. What is SQL?*

**English answer:** SQL stands for Structured Query Language. It is used to store, retrieve, manipulate, and manage data in relational databases.

**Hindi explanation:** SQL ka full form Structured Query Language hai. Isse relational database me data read, insert, update, delete aur manage karte hain.

**Example:** `Example: SELECT name, salary FROM employees;`

### *Q2. What is a database?*

**English answer:** A database is an organized collection of data. A DBMS is software used to create and manage databases; examples include MySQL, SQL Server, and PostgreSQL.

**Hindi explanation:** Database organized data ka collection hota hai. DBMS us data ko manage karne wala software hai.

### *Q3. What is a table, row, and column?*

**English answer:** A table stores related data. A row is one record, and a column is one attribute or field.

**Hindi explanation:** Table data ko rows aur columns me rakhti hai. Row ek record aur column ek property hoti hai.

### *Q4. What is a primary key?*

**English answer:** A primary key uniquely identifies each row. It cannot contain duplicate or NULL values.

**Hindi explanation:** Primary key har row ki unique identity hoti hai; duplicate aur NULL allowed nahi hote.

### *Q5. What is a foreign key?*

**English answer:** A foreign key links a column in one table to a key in another table and helps maintain referential integrity.

**Hindi explanation:** Foreign key do tables ke beech relationship banati hai aur valid references maintain karti hai.

### *Q6. WHERE vs HAVING?*

**English answer:** WHERE filters rows before grouping; HAVING filters groups after GROUP BY.

**Hindi explanation:** WHERE grouping se pehle rows filter karta hai; HAVING aggregate/group banne ke baad groups filter karta hai.

**Example:** `SELECT department, AVG(salary) FROM employees GROUP BY department HAVING AVG(salary)>3000;`

### *Q7. What is GROUP BY?*

**English answer:** GROUP BY combines rows with the same values so aggregate functions can be calculated per group.

**Hindi explanation:** GROUP BY same value wali rows ko group karta hai, jaise department-wise salary average.

### *Q8. What are aggregate functions?*

**English answer:** COUNT, SUM, AVG, MIN, and MAX summarize multiple rows into a value.

**Hindi explanation:** Ye functions multiple rows ka summary dete hain—count, total, average, minimum aur maximum.

### *Q9. What is JOIN?*

**English answer:** A JOIN combines related rows from two or more tables using a matching condition.

**Hindi explanation:** JOIN common key ke basis par tables ka related data combine karta hai.

### *Q10. Explain INNER JOIN.*

**English answer:** INNER JOIN returns only rows with matching keys in both tables.

**Hindi explanation:** Dono tables me matching key wali rows hi aati hain.

**Example:** `SELECT * FROM Orders o INNER JOIN Customers c ON o.CustomerID=c.CustomerID;`

### *Q11. LEFT JOIN vs RIGHT JOIN?*

**English answer:** LEFT JOIN keeps every row from the left table and matching right rows; RIGHT JOIN does the reverse.

**Hindi explanation:** LEFT JOIN left table ki saari rows rakhta hai; RIGHT JOIN right table ki saari rows rakhta hai. Match na ho to doosri side NULL hoti hai.

### *Q12. What is FULL OUTER JOIN?*

**English answer:** It returns matched rows plus unmatched rows from both tables, where supported by the database.

**Hindi explanation:** Dono tables ki matching aur non-matching rows return hoti hain.

### *Q13. What is a self join?*

**English answer:** A self join joins a table to itself, often to connect employees with their managers.

**Hindi explanation:** Ek hi table ko khud ke saath join karna, jaise employee-manager mapping.

### *Q14. UNION vs UNION ALL?*

**English answer:** UNION removes duplicate rows; UNION ALL keeps duplicates and is usually faster.

**Hindi explanation:** UNION duplicates hata deta hai; UNION ALL duplicates rakhta hai.

### *Q15. What is a subquery?*

**English answer:** A subquery is a query nested inside another SQL statement.

**Hindi explanation:** Ek query ke andar likhi hui query subquery hoti hai.

**Example:** `SELECT name FROM employees WHERE salary>(SELECT AVG(salary) FROM employees);`

### *Q16. What is a CTE?*

**English answer:** A Common Table Expression is a named temporary result set defined with WITH, improving readability of a query.

**Hindi explanation:** WITH se banaya gaya named temporary result CTE hai; complex query readable banti hai.

**Example:** `WITH high_paid AS (SELECT * FROM employees WHERE salary>5000) SELECT * FROM high_paid;`

### *Q17. What is a window function?*

**English answer:** A window function calculates across related rows without collapsing them into one row per group.

**Hindi explanation:** Window function calculation karti hai lekin original rows ko collapse nahi karti.

**Example:** `SELECT name, salary, RANK() OVER(ORDER BY salary DESC) AS rnk FROM employees;`

### *Q18. ROW_NUMBER vs RANK vs DENSE_RANK?*

**English answer:** ROW_NUMBER assigns unique sequence numbers; RANK leaves gaps after ties; DENSE_RANK does not leave gaps.

**Hindi explanation:** Tie hone par ROW_NUMBER unique number deta hai; RANK me gap aata hai; DENSE_RANK me gap nahi.

### *Q19. How do you find the second-highest salary?*

**English answer:** Use DENSE_RANK or a distinct ordered subquery, depending on whether ties should count as one rank.

**Hindi explanation:** Second distinct salary ke liye DENSE_RANK use kar sakte hain; tie ko same rank milta hai.

**Example:** `SELECT salary FROM (SELECT salary,DENSE_RANK() OVER(ORDER BY salary DESC) r FROM employees) x WHERE r=2;`

### *Q20. How do you find employees earning above 3000?*

**English answer:** Filter with WHERE and select only the requested columns.

**Hindi explanation:** WHERE se salary condition lagao aur sirf name aur salary select karo.

**Example:** `SELECT employee_name, salary FROM employees WHERE salary>3000;`

### *Q21. How do you find names ending with 'a'?*

**English answer:** Use LIKE with a trailing-character pattern.

**Hindi explanation:** Last character 'a' ho to LIKE '%a' use karte hain.

**Example:** `SELECT name FROM employees WHERE name LIKE '%a';`

### *Q22. What is NULL?*

**English answer:** NULL represents missing or unknown data. Use IS NULL or IS NOT NULL, not = NULL.

**Hindi explanation:** NULL missing/unknown value hai. Iske liye IS NULL likhte hain, = NULL nahi.

### *Q23. DELETE vs TRUNCATE vs DROP?*

**English answer:** DELETE removes selected rows and can use WHERE; TRUNCATE removes all rows efficiently; DROP removes the table object.

**Hindi explanation:** DELETE selected rows hata sakta hai; TRUNCATE table ka data clear karta hai; DROP table hi remove karta hai. Transaction behavior DBMS par depend karta hai.

### *Q24. What is normalization?*

**English answer:** Normalization organizes data to reduce duplication and update anomalies, commonly through 1NF, 2NF, and 3NF.

**Hindi explanation:** Normalization duplicate data aur update problems kam karne ke liye tables ko structure karna hai.

### *Q25. What is an index?*

**English answer:** An index can speed up data retrieval but consumes storage and can slow writes.

**Hindi explanation:** Index searching fast kar sakta hai, lekin storage leta hai aur insert/update ko slow kar sakta hai.

### *Q26. What is a view?*

**English answer:** A view is a saved query presented like a virtual table; it can simplify access to selected data.

**Hindi explanation:** View saved SELECT query jaisa virtual table hota hai.

### *Q27. What is a stored procedure?*

**English answer:** A stored procedure is a named set of SQL statements stored in the database and executed when called.

**Hindi explanation:** Database me saved SQL statements ka reusable group stored procedure hai.

### *Q28. What is a transaction?*

**English answer:** A transaction is a unit of work governed by ACID properties: atomicity, consistency, isolation, durability.

**Hindi explanation:** Transaction ek logical work unit hai. ACID data reliability ke principles hain.

### *Q29. How do you handle duplicate records?*

**English answer:** Identify duplicates with GROUP BY/HAVING or ROW_NUMBER, then decide which record to retain before deleting.

**Hindi explanation:** Pehle duplicates identify karo, business rule se decide karo kaunsa record rakhna hai, phir safely remove karo.

### *Q30. How do you optimize a slow query?*

**English answer:** Check the execution plan, indexes, filters, join keys, selected columns, data types, and row volume; measure changes.

**Hindi explanation:** Execution plan dekho, unnecessary columns avoid karo, joins/indexes aur filters check karo; change ko measure karo.

### *Q31. What is CASE WHEN?*

**English answer:** CASE adds conditional logic to a query, similar to if-else.

**Hindi explanation:** SQL me condition ke basis par category banane ke liye CASE use hota hai.

**Example:** `SELECT name, CASE WHEN salary>=5000 THEN 'High' ELSE 'Standard' END AS band FROM employees;`

### *Q32. What is COALESCE?*

**English answer:** COALESCE returns the first non-NULL expression in its argument list.

**Hindi explanation:** Multiple values me pehli non-NULL value return karta hai.

**Example:** `SELECT COALESCE(phone,'Not Available') FROM customers;`

### *Q33. What is the difference between CHAR and VARCHAR?*

**English answer:** CHAR is fixed-length; VARCHAR stores variable-length text, subject to DBMS behavior.

**Hindi explanation:** CHAR fixed length aur VARCHAR variable length text ke liye hota hai.

### *Q34. What is a date function use case?*

**English answer:** Date functions support month-wise reporting, date filtering, and extracting year/month components; syntax varies by SQL engine.

**Hindi explanation:** Date functions se month-wise sales, year/month extraction aur date filters karte hain. Syntax database ke hisab se badal sakta hai.

### *Q35. How would you calculate monthly sales?*

**English answer:** Group order totals by year and month, ensuring the date expression matches your SQL dialect.

**Hindi explanation:** Order date ke year-month par group karke SUM(sales) nikaalenge.

### *Q36. What is a data analyst's SQL workflow?*

**English answer:** Understand the business question, inspect schema, validate row counts and nulls, query in stages, reconcile totals, and document assumptions.

**Hindi explanation:** Pehle business question samjho, schema dekho, null/duplicates check karo, query likho aur totals validate karo.

### *Q37. How do you explain SQL in a project?*

**English answer:** I used SQL to extract relevant records, join tables, clean/filter data, aggregate KPIs, and validate results before visualization.

**Hindi explanation:** Project me SQL se data extract, tables join, filtering, KPI aggregation aur Power BI se pehle validation ki.

### *Q38. What is the difference between OLTP and OLAP?*

**English answer:** OLTP supports frequent operational transactions; OLAP supports analytical queries over larger historical datasets.

**Hindi explanation:** OLTP daily transactions ke liye; OLAP analysis aur reporting ke liye hota hai.

### *Q39. What is a constraint?*

**English answer:** Constraints enforce rules such as NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY, and CHECK.

**Hindi explanation:** Constraints data par rules lagate hain, jaise NOT NULL, UNIQUE aur CHECK.

### *Q40. How do you safely update records?*

**English answer:** Preview rows with SELECT and the same WHERE condition, use a transaction when appropriate, then verify affected rows.

**Hindi explanation:** UPDATE se pehle same WHERE ke saath SELECT karke rows verify karo; phir update aur result check karo.

Interview tip: If you are unsure of a detail, explain your understanding honestly and describe how you would verify it.